

ROCKGLEN, SK, Canada

WMO: 711350

Lat: 49.1678N

Lon: 105.9792W

Elev: 917

StdP: 90.78

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-26.2	-33.6	0.2	-29.2	-30.1	0.3	-26.1	14.0	-4.6	12.8	-5.4	4.1	320	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	31.1	17.0	29.1	16.5	27.1	15.9	19.2	26.6	18.1	25.4	17.1	24.5	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.5	22.0	15.6	12.4	20.6	14.4	11.5	19.8	58.8	26.9	54.7	25.2	51.5	24.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.4	9.4	DB	-32.8	34.5	3.0	2.0	-34.9	35.9	-36.6	37.1	-38.3	38.2	-40.5	39.7	(4)
(5)			WB	-32.7	21.5	3.1	1.5	-34.9	22.6	-36.7	23.4	-38.4	24.3	-40.6	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-10.6	-9.4	-3.5	4.2	10.3	14.9	19.0	18.4	13.1	5.1	-2.7	-9.0	(6)
(7)		DBStd	12.20	8.44	7.76	7.61	5.69	4.66	3.62	3.38	3.89	5.08	5.54	7.03	7.69	(7)
(8)		HDD10.0	3008	640	543	418	185	53	4	0	1	24	167	383	589	(8)
(9)		HDD18.3	5279	898	777	677	425	252	112	32	49	169	409	632	847	(9)
(10)		CDD10.0	901	0	0	1	11	62	153	281	259	118	16	1	0	(10)
(11)		CDD18.3	129	0	0	0	0	2	10	54	50	12	1	0	0	(11)
(12)		CDH23.3	1578	0	0	0	4	42	132	620	597	177	6	0	0	(12)
(13)		CDH26.7	513	0	0	0	1	5	27	216	213	52	1	0	0	(13)
(14)	Wind	WSAvg	4.7	5.2	4.9	5.0	4.9	4.7	4.2	4.0	4.3	5.0	5.1	5.3	(14)	
(15)	Precipitation	PrecAvg	356	16	10	17	27	56	65	54	38	26	23	13	17	(15)
(16)		PrecMax	544	45	20	45	57	188	140	214	107	96	100	51	48	(16)
(17)		PrecMin	220	3	1	1	6	9	1	16	3	3	2	0	2	(17)
(18)		PrecStd	72	10	6	11	14	43	28	40	24	20	22	11	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	7.7	16.7	23.3	27.1	30.0	34.5	34.2	31.3	23.8	16.6	7.2	(19)
(20)			MCWB	2.7	3.8	8.4	10.9	13.9	16.9	17.7	17.5	16.1	12.0	8.4	2.9	(20)
(21)		2%	DB	3.8	4.5	12.0	19.2	24.2	26.7	31.2	31.3	27.9	19.2	11.8	4.6	(21)
(22)			MCWB	1.3	1.8	5.6	9.5	13.3	16.6	17.7	16.6	14.4	10.2	6.1	1.5	(22)
(23)		5%	DB	2.4	2.7	8.5	16.2	21.6	24.6	29.1	29.0	25.1	16.2	8.6	2.7	(23)
(24)			MCWB	0.4	0.5	3.9	7.9	11.9	15.4	17.3	16.1	13.9	9.0	4.0	0.4	(24)
(25)		10%	DB	0.6	0.6	5.6	13.5	19.0	22.5	26.9	26.9	22.1	13.3	6.0	0.8	(25)
(26)			MCWB	-1.0	-0.9	2.4	6.6	10.5	14.4	16.7	15.6	12.5	7.7	2.6	-1.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	4.2	8.4	12.0	15.6	20.0	21.4	19.7	17.4	13.2	9.0	3.3	(27)
(28)			MCDWB	5.5	7.2	15.9	21.4	23.5	26.1	28.0	28.2	26.9	21.4	15.9	6.2	(28)
(29)		2%	WB	1.5	2.1	6.0	10.0	13.9	17.7	19.9	18.3	15.7	11.1	6.3	1.7	(29)
(30)			MCDWB	3.5	4.1	11.0	18.3	21.9	24.1	27.2	26.8	24.7	17.8	11.2	4.2	(30)
(31)		5%	WB	0.5	0.6	4.1	8.5	12.7	16.4	18.7	17.3	14.3	9.5	4.3	0.4	(31)
(32)			MCDWB	2.3	2.4	7.9	15.2	20.0	22.6	25.7	25.7	23.0	15.3	8.1	2.6	(32)
(33)		10%	WB	-1.0	-0.9	2.5	7.0	11.5	15.2	17.6	16.4	13.1	8.0	2.7	-0.9	(33)
(34)			MCDWB	0.6	0.6	5.4	12.8	17.7	21.2	24.6	24.3	21.3	13.0	5.6	0.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	8.7	9.2	11.4	11.8	11.0	12.7	13.0	12.2	10.4	8.9	8.1	(35)
(36)			MCDBR	9.5	9.4	13.1	16.0	15.7	14.6	16.0	16.3	16.7	15.1	12.3	9.2	(36)
(37)			MCWBR	7.6	7.1	8.0	8.6	7.8	6.9	6.2	6.1	7.1	7.9	7.4	6.9	(37)
(38)		5% WB	MCDBR	9.3	9.2	12.5	14.8	13.9	12.8	13.5	14.2	14.9	13.9	12.0	9.0	(38)
(39)			MCWBR	7.7	7.3	8.0	8.3	7.3	6.8	6.4	6.1	6.9	7.7	7.6	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.251	0.267	0.316	0.337	0.339	0.351	0.359	0.317	0.277	0.250	0.229	(40)	
(41)		taud	2.431	2.469	2.545	2.412	2.396	2.440	2.423	2.372	2.460	2.563	2.529	2.487	(41)	
(42)		Ebn at Noon	842	925	964	934	922	919	902	877	888	882	838	827	(42)	
(43)		Edh at Noon	61	75	83	106	113	109	109	110	90	68	55	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.35	2.37	3.65	4.90	5.85	6.30	6.82	5.54	4.05	2.53	1.46	1.04	(44)	
(45)		RadStd	0.12	0.16	0.22	0.28	0.43	0.38	0.37	0.39	0.36	0.27	0.17	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page